

AI Security Lab — Level 01: Say Hello

Vulnerability Overview

- **Vulnerability Type:** Application Logic / Multi-Turn Interaction
- **Objective:** Interact with the AI assistant across multiple messages and evaluate how application-level state and interaction counters affect challenge behavior.

Technical Analysis

This laboratory demonstrates that an AI-powered application is not composed solely of the underlying language model. The surrounding application typically contains **orchestration logic, session management, interaction counters, evaluation rules, and state-management mechanisms**.

The application monitors the number and sequence of interactions and uses this state as part of its evaluation logic.

This highlights the importance of assessing not only the LLM itself, but also the application logic surrounding the model.

Security Impact

Weak session-state management or flawed interaction logic may result in:

- Incorrect challenge-state evaluation.
- Manipulation of application workflows.
- Inconsistent authorization or scoring behavior.
- State-related logic bypasses.

Remediation

- Implement robust **server-side session-state tracking**.
- Validate interaction state independently on the server.
- Ensure counters and evaluation logic cannot be manipulated through client-controlled values.
- Apply deterministic validation to multi-turn workflows.
- Maintain clear separation between model-generated content and application-control logic.

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